

Kryptos K2/K4 Internal Variant System

Proposed Method That Was Used to Build K4

- Start with by populating a 26 x 26 matrix of character lookups with, say 12.6% E's e.g. 26x26x0.126 --> 85 or even those crypto FM phrases like SIXTYSEVEN, LUCKYCHARM, MASTERWORK etc... (this hides the English frequencies)
- Using K2 as your random X reference character ("Q") find one of your K4 PT character in that same row ("I") and pick its Y column reference ("T")
- "T" then becomes your K4 CT for that position
- If you start seeing too many bigrams/trigrams, go back and choose an alternative letter from the original selections.
- Publish the string of Y references as the CT code
- Add four characters to the front of the CT code (modified with ROT-13) to indicated where to find the other half of the bigram OBKR/BOXE

1st Steps Working Backwards to Solve

- start with an empty 26 x 26 matrix for the bigram characters (different row/column counts could apply but we can compress/collapse later)
- fill in the known cribs/letters into the array
- regenerate all of K4 with the new matrix

Example

K2

ELZZVRRGKFFV0EEXBDMVPNFQXEZLGRE
DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP
DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG

K4

U0X0GHULBSOLIFBBWFLRVQQPRNGKSS0
TWTQJSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDKZXTJCIDIGKUHUAUEKCAR

So "QT" corresponds to the "I" in the known crib "BERLIN" (Line 3/2nd letter from the left)

Our internal variant matrix would therefore assign "I" to the **QT** reference

Repeating this for all known CT/PT clues a regenerating the whole shebang from the start yeilds:

```

.....EASTNORTHEAST.
..IO..N.S.....E.BER
LINCLOCK.....A.....

```

The word preceding BERLIN without the ".E." hint would normally be assumed to be "THE". We can now assume this is not the case.

Further Steps

Based on the context of Kryptos we can start to look at stuff related to the BERLIN CLOCK, its underground station, maybe the poet/his works that the station is named for (Uhland) or even the fact it was at the initial East/West divide when the wall went up. Phrases like "UNDER berlin clock's terminus/station..." and "BELOW Berlin clock's shadow" seem to fit so we'll a few...

Using the word "UNDER" immediately prior to "BERLIN" then check a few things:

1. That we do not clash with already allocated letters within the same matrix reference i.e. we do not want anything but an "I" for "QT"
2. Try fitting words/phrases in around the new "anchoring" letters

Here I've highlighted the additional letters in BOLD when trialling "UNDER"

```

.....R.EASTNORTHEAST.
..IO..N.S.....UNDERBER
LINCLOCK.....A.....

```

The word after ENE seems to be place related and "REGION" seems to fit the bill.

Now this combo result in the matrix having the same letter "E" on the same 4th or "D" row . We'll ignore that because we don't mind lots of duplicate letters for frequency masking.

So now we've got the following:

```

.....R.EASTNORTHEASTR
EGION.N.S....S.....UNDERBER
LINCLOCKS.....A.....

```

There are no new insights/"anchoring" letters but then again, there are no clashes, so it is a valid combo.

As is this:

```

.....R.EASTNORTHEASTR
EGIONANDSEEKSTOREUNIFIYUNDERBER
LINCLOCKT.....R.....A.....IE.....

```

Matrix Analysis

So far the filled in matrix yields no obvious insights as to the method used to fill in the initial matrix - at least in its generated form.

Further analysis manipulations, alphabet shuffling with keywords need to be applied to see if anything comes to light e.g. rearrange the internal rows alphabetically so see if the column header/reference spells anything interesting...

```

    ['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K',
'L', 'M', 'N', 'O', 'P', 'Q', 'R', 'S', 'T', 'U', 'V',
'W', 'X', 'Y', 'Z']
A ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'N', '.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
B ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
C ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
D ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
E ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', 'H', 'R', 'C', '.', '.', '.', '.', '.',
'.', '.', '.', 'L']
F ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', 'O', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
G ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', 'S', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
H ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
I ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', '.']
J ['.', '.', '.', '.', '.', '.', '.', '.', '.', '.',
'.', '.', '.', '.', 'R', '.', '.', '.', '.',
'.', '.', '.', '.', '.']

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[illegible]

Alternative Attack Methods

- Perhaps standard biliteral systems attacks (reference: Page 92, Section 5-10 of the Basic Cryptanalysis Army Field manual)
- Dictionary fillers
- AI/ML predict previous word in phrase/predict subsequent word in phrase (tensorflow/BERT models/seq2seq)
- ??
- ??

Pros

- Pen and Paper style with a little programming help with bigram matrix generation/checking.
- "as simple as ABC" - where you have x="A", y="B" and the lookup is "C"
- "is in plain sight" - the arrangement of the last three lines of K2 and it's corresponding shape in K4 indicate a 1:1 relationship of some sort. The XY lookup has been used already in K1/K2 with Vigenere
- internal variant system - hides the frequencies used automatically - no binaries, ones, zeros though, entirely human/analogue - and hard
- having the same letters enciphered by multiple bigrams make it easier to ensure you can easily put in clues/breadcrumbs of how to solve e.g. TUT,WIZ,KID,PKS,ODD,etc
- the same bigram freedom can also ensure the encoder has multiple selections to choose from so as to avoid (after splitting the second letter off into K4) English bigrams/trigrams
- BOX E can (at a stretch) be loosely identified with last 3 lines of K2/ certainly the last 2.
- Can be rotated every odd lookup
- the same degrees of freedom likewise allows you the freedom to assign a secondary key-phrase to create a duress cipher
- Can be rotated every odd lookup - Not sure this "washing machine" is truly K4 or just K3 where it applies 100%...
- impossible to solve with only the K4 half of the CT in play :)

Cons

- It extends itself beyond K4 proper - Is this still fair?
- It is a bigram system 97 and also 93 are not divisible by 2, so technically K4 in itself is not sufficient to solve K4
- The encipherment is only represented by 50% of K4 - the other 50% is in bottom bit of K2
- The placement of the lookup characters in the internal variant matrix (using A thru Z & A thru Z) don't yield any patterns to suggest how to fill in the remaining missing letters - yet!
- a 1:1 between K4's CT and K4's PT isn't strictly in play due to the other half

of K4's CT existing over in K2 - the relative positions of NYPTT/BERLIN remain intact

- could be the final masking solution of an interim K4 CT, so the method of attack is not strictly the right way to approach this....